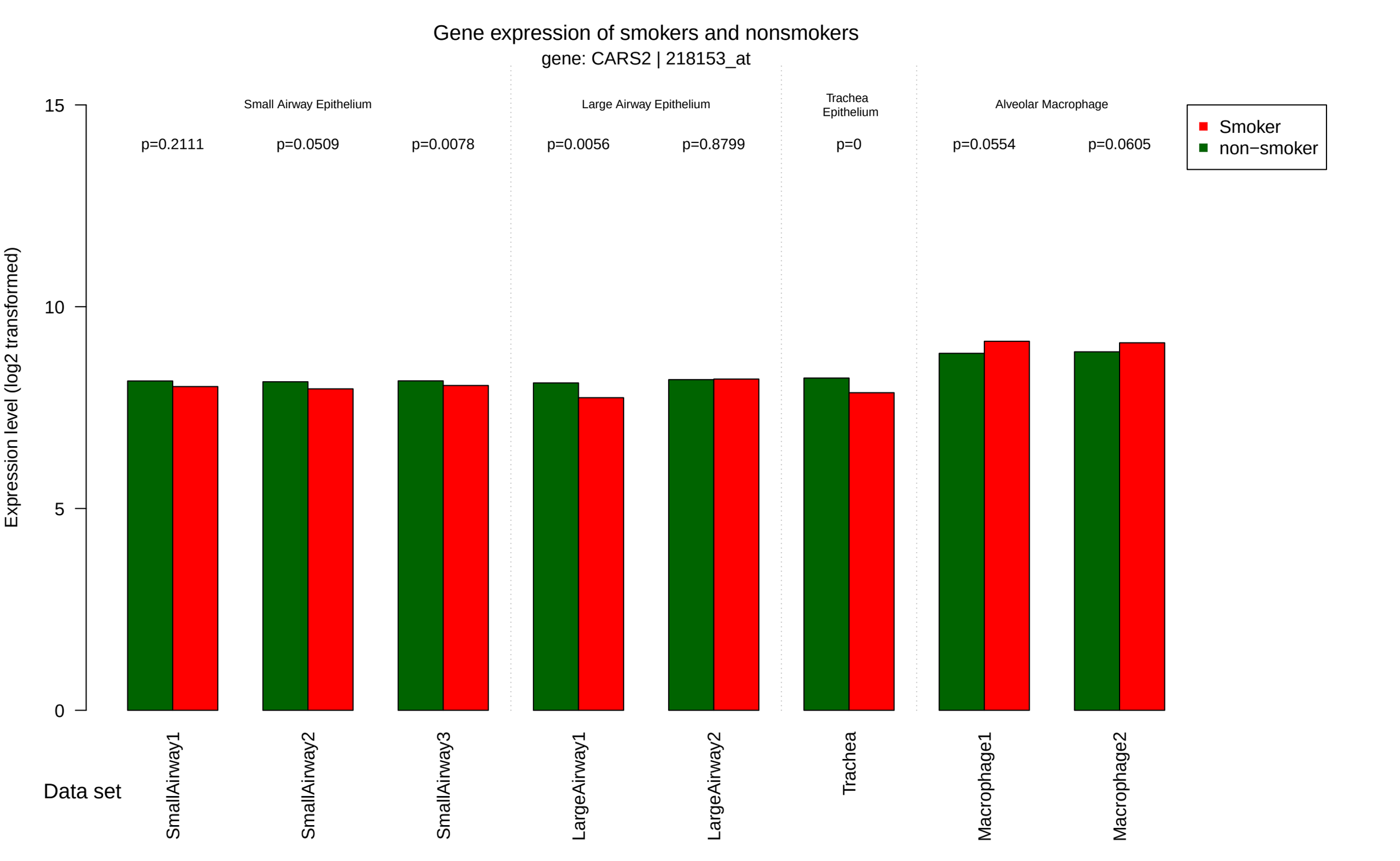
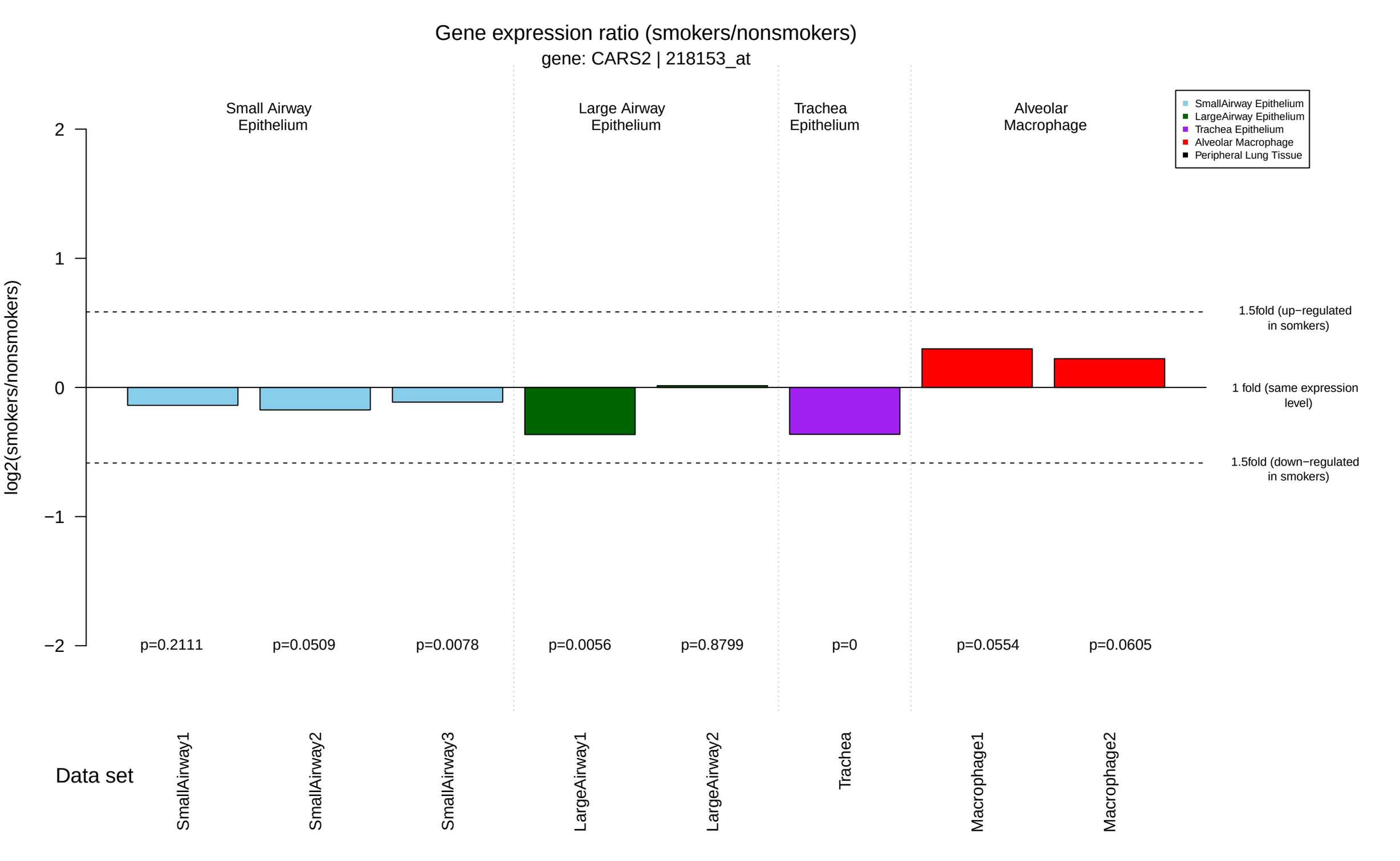
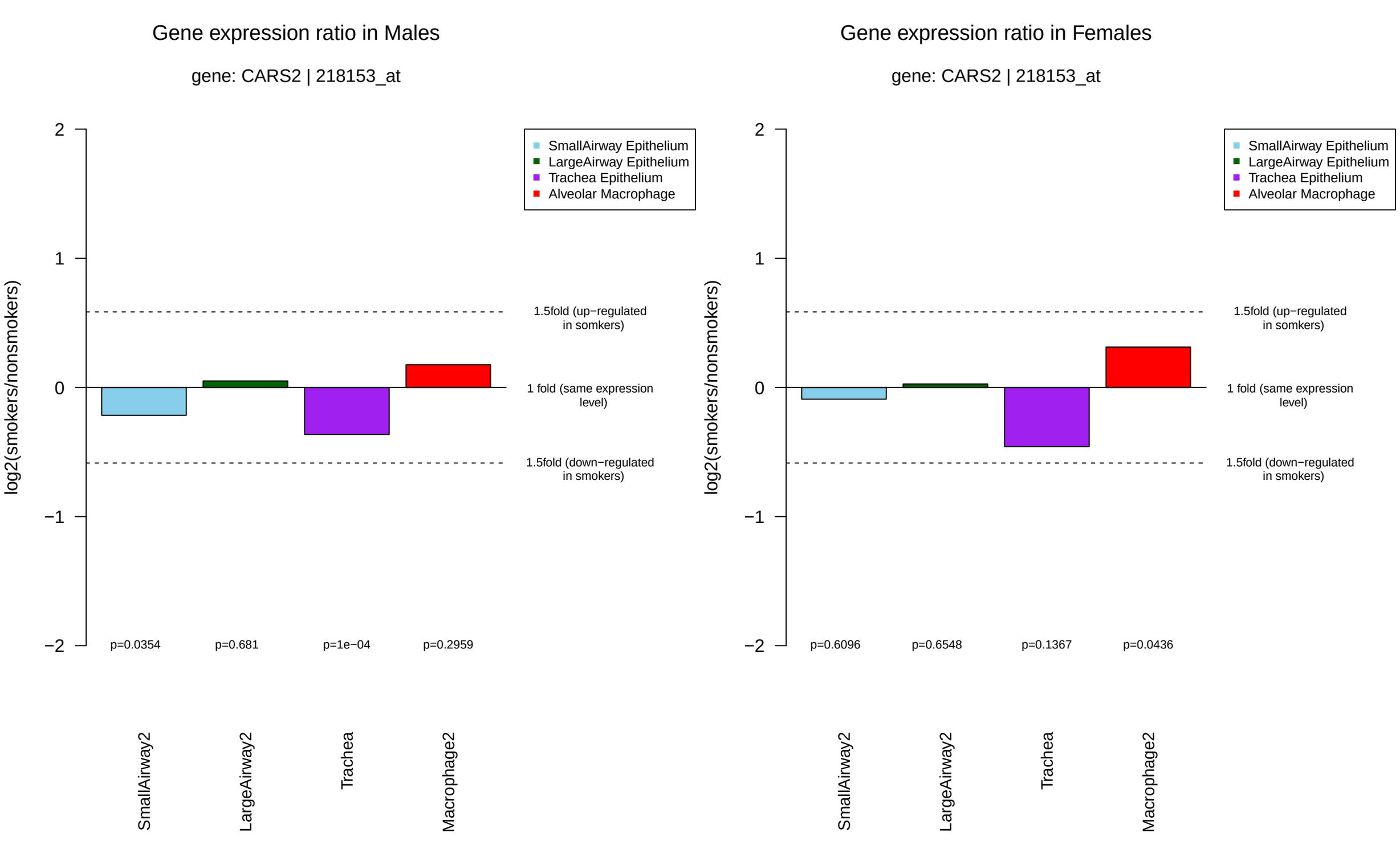
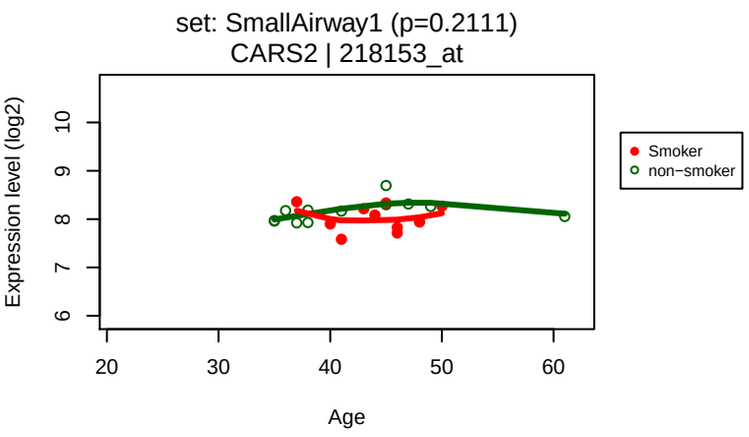
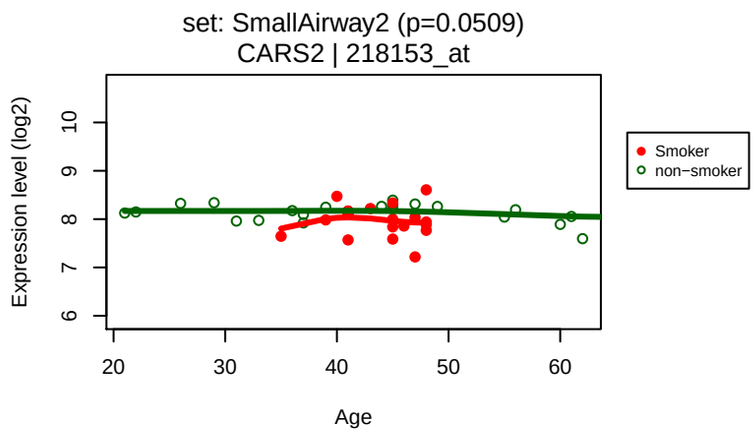
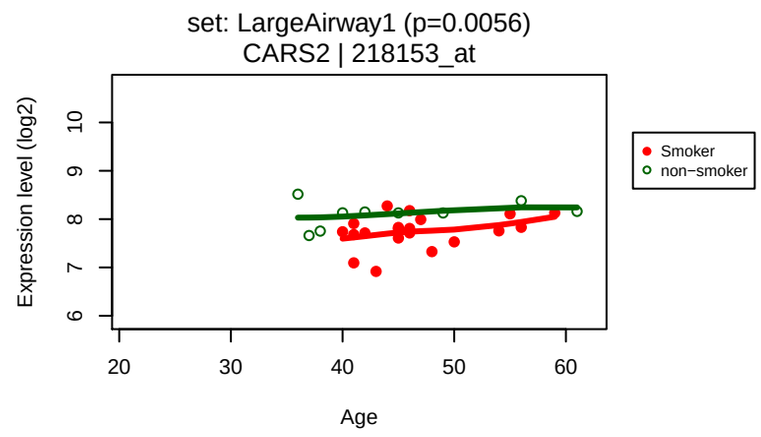

Gene expression of smokers and nonsmokers
gene: CARS2 | 218153_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2111
p=0.0509
p=0.0078
p=0.0056
p=0.8799
p=0
p=0.0554
p=0.0605
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

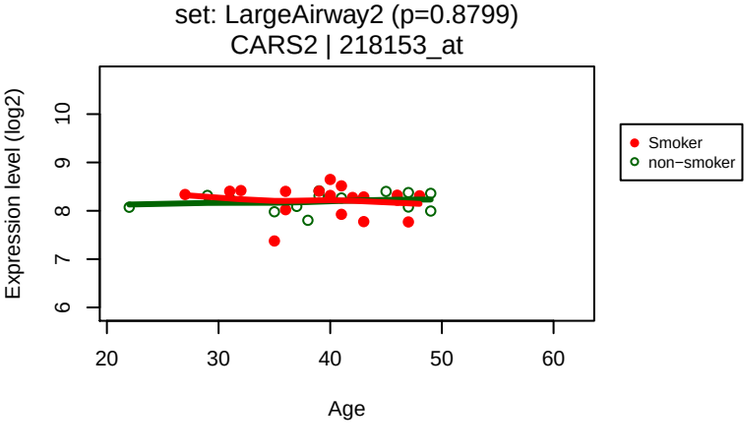

Gene expression ratio (smokers/nonsmokers)
gene: CARS2 | 218153_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2111
p=0.0509
p=0.0078
p=0.0056
p=0.8799
p=0
p=0.0554
p=0.0605
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

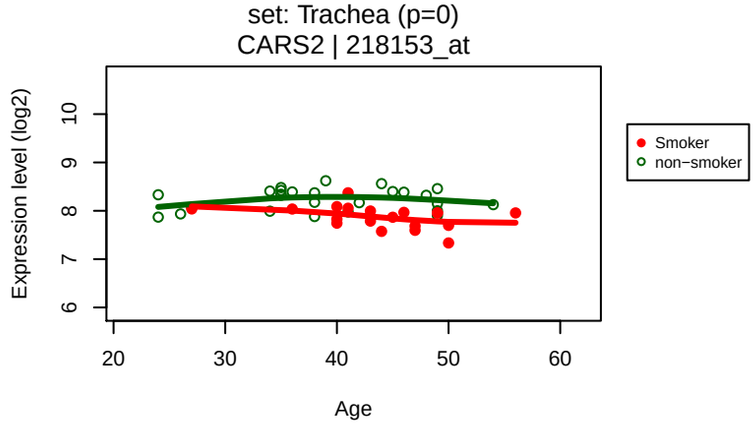

Gene expression ratio in Males
gene: CARS2 | 218153_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0354
p=0.681
p=1e−04
p=0.2959
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CARS2 | 218153_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6096
p=0.6548
p=0.1367
p=0.0436
SmallAirway2
LargeAirway2
Trachea
Macrophage2

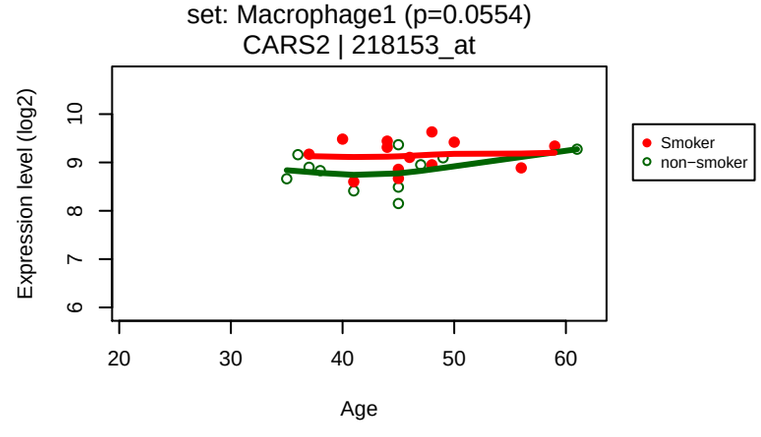

set: SmallAirway1 (p=0.2111)
CARS2 | 218153_at
Expression level (log2)
Age
Smoker
non−smoker

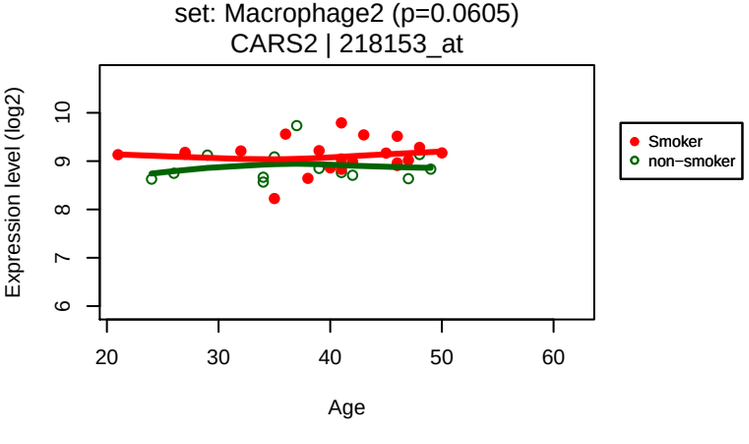

set: SmallAirway2 (p=0.0509)
CARS2 | 218153_at
Expression level (log2)
Age
Smoker
non−smoker

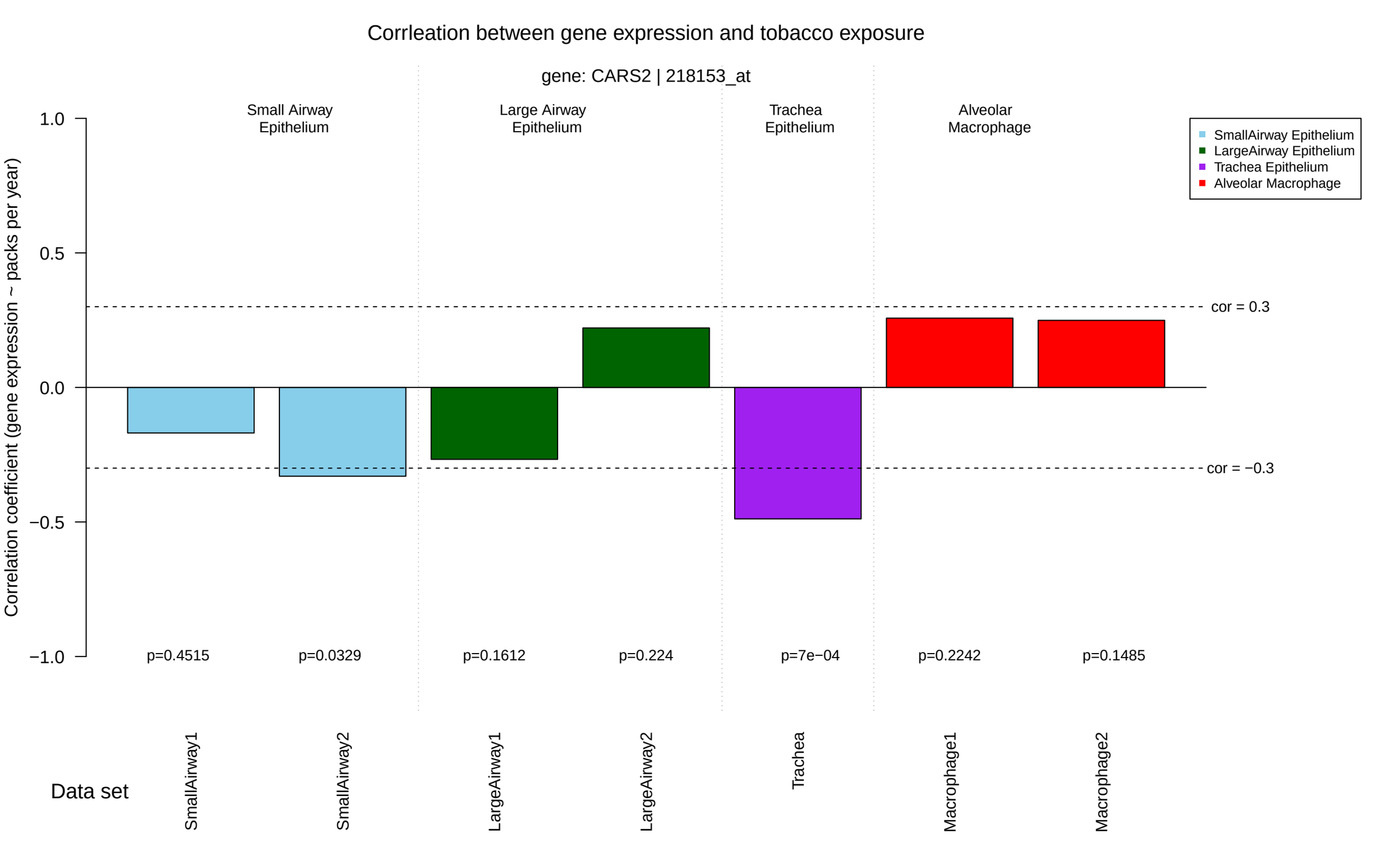

set: LargeAirway1 (p=0.0056)
CARS2 | 218153_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8799)
CARS2 | 218153_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
CARS2 | 218153_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0554)
CARS2 | 218153_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0605)
CARS2 | 218153_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CARS2 | 218153_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.4515
p=0.0329
p=0.1612
p=0.224
p=7e−04
p=0.2242
p=0.1485
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2